Math Class 9th Federal Board Chapter 05 Linear Equations and Inequalities Full Test

1 MCQ
1 SQ } 5 marks

Content Domain / Area	Topics	SLO No./ Description	Form of Assessment	Cognitive Level (Knowledge, Understanding, Application)	Remarks
Linear Equations and Inequalities in one variable		**[SLO: M-09-A-22]:** Solve linear equations and inequalities with rational coefficients and represent the solution set on a real line.	Summative	U	**Question(s) will be asked in the annual examination**
		[SLO: M-10-A-23]: Solve two linear inequalities with two unknowns simultaneously	Formative	U	**Question(s) will not be asked in the annual examination (This SLO will be taught in Class X)**

Q.1 Choose the correct option.

$\frac{-2x}{-2} \geq \frac{4}{-2} \Rightarrow x \geq -2$

1. $-2x \leq 4$ gives:

   (A) $x \leq -2$ (B) $x \geq -2$ (C) $x \leq -1$ (D) $x = -1$

2. $x + 3 > 5$ implies:

   $x > 5 - 3 \Rightarrow x > 2$

   (A) $x = 0$ (B) $x < 2$ (C) $x \leq 2$ (D) $x > 2$

3. Solution of the inequality $-2x - \frac{1}{2} \leq \frac{3}{2}$ is:

   (A) $x > -1$ (B) $x < -1$ (C) $x \geq -1$ (D) $x \leq -1$

$<$ less than

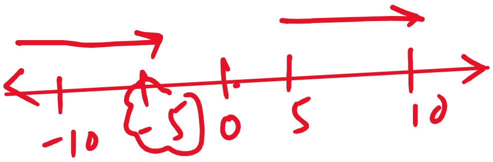
$\leq$ less than or equal to

$-2x \leq 4$

$5 \leq 10$

$\downarrow$

$(-5) \geq (-10)$



$\frac{1}{1} < \frac{10}{1}$

$1 > \frac{1}{10}$

3. Solution of the inequality $-2x - \frac{1}{2} \leq \frac{3}{2}$ is:

   (A) $x > -1$ (B) $x < -1$ (C) $x \geq -1$ (D) $x \leq -1$

3. Solution of the inequality $-2x - \frac{1}{2} \leq \frac{3}{2}$ is:

(A) $x > -1$ (B) $x < -1$ (C) $x \geq -1$ (D) $x \leq -1$

$-2x \leq \frac{3}{2} + \frac{1}{2}$ $\frac{3+1}{2} = \frac{4}{2} = 2$

$-2x \leq 2$

$\frac{\cancel{-2}x}{\cancel{-2}} \geq \frac{2}{-2} \Rightarrow x \geq -1$

4. A pizza costs Rs. 1350, and you have Rs. 2500. The maximum amount of pizza that can be bought is represented by:

(A) $x \geq 1.85$ (B) $x \leq 1.85$ (C) $x > 1.85$ (D) $x = 2$

$x \quad \frac{2500}{1350}$ $\frac{500}{100} = 5$

$x \leq 1.85$

$1350x \leq 2500$

$x \leq \frac{2500}{1350} \leq 1.85$

5. The solution to the inequality $x + 3 < 5$ is:

(A) $x > 2$ (B) $x < 2$ (C) $x = 2$ (D) $x \geq 2$

6. If x is smaller than 10, then which statement is true?

(A) $x \geq 20$ (B) $x \leq 10$ (C) $x < 10$ (D) $x > 10$

less $x < 5 - 3 \Rightarrow x < 2$

$-\frac{7}{2} = (-3.5)$ $x < 10$ $x \geq -2$

$-\frac{7}{2}=$ $x < 10$

$x \le -2$

7. Which of the following is a solution of the inequality $3 - 4x \le 11$?

(A) -8 (B) -2 (C) $-\frac{14}{4}$ (D) None of these

8. $x =$ ____ is a solution of the inequality $-2 < x < \frac{3}{2}$. $\Rightarrow -2 < x < 1.5$

(A) -5 (B) 3 (C) 0 (D) $\frac{3}{2}$

9. If x is no larger than 10, then:

(A) $x \ge 8$ (B) $x \le 10$ (C) $x < 10$ (D) $x > 10$

9) $x \le 10$

$3 - 4x \le 11$

$-4x \le 11 - 3$

$-4x \le 8$

$\frac{-4x}{-4} \ge \frac{8}{-4}$

$\boxed{x \ge -2}$

$[-2, +\infty)$

0

10. If the capacity c of an elevator is at most 1600 pounds, then:

(A) $c < 1600$ (B) $c \ge 1600$ (C) $c \le 1600$ (D) $c > 1600$

$c \le 1600$

at least $\ge$

$5 < 0$ $2 < 0$ $-2 < 0$

11. $x = 0$ is a solution of the inequality:

(A) $x > 0$ (B) $3x + 5 < 0$ (C) $x + 2 < 0$ (D) $x - 2 < 0$

$-2 < 0$

11. $x = 0$ is a solution of the inequality:

(A) $x > 0$ (B) $3x + 5 < 0$ (C) $x + 2 < 0$ (D) $x - 2 < 0$

12. Which one is the standard form of a linear equation?

(A) $ax^2 + b = 0$ (B) $ax + b = -c$ (C) $ax + b > 0$ (D) $ax + b = 0$

13. What number must be subtracted from the right side of $7x = 30$ so that 4 is a solution of the resulting equation?

(A) 7 (B) 2 (C) 4 (D) -2

RHS

$\boxed{x = 4}$

$-2 \quad 0$

$7x = 30$

$7x = 30 - 2 = 28$

$x = \frac{28}{7} = 4$

14. Which one is the solution of $\frac{4}{x} - \frac{2}{x} = 5$?

(A) -1 (B) $\frac{2}{5}$ (C) $\frac{5}{2}$ (D) zero

15. Which one is a strict inequality?

(A) $x + 3 \neq 0$ (B) $12x > 5$ (C) $2y - 3 \leq 0$ (D) $4x + 5 \geq 0$

$\frac{4-2}{x} = 5$

$\Rightarrow \frac{2}{x} = \frac{5}{1}$

$\frac{x}{2} = \frac{1}{5}$

$x = 2/5$

$5x = 2$

$\Rightarrow x = \frac{2}{5}$

$2x + 12 \leq 3x + 12 \Rightarrow 2x - 3x \leq 0$

$-x \leq 0$

$\Downarrow$

16. Which one is the solution of $2(x + 6) \leq 3(x + 4)$?

(A) $x \leq 0$ (B) $x \geq 0$ (C) $x \leq 24$ (D) $x \geq 24$

$2x + \cancel{12} \leq 3x + \cancel{12}$

$x \geq 0$

$$2x + \cancel{12} \le 3x + \cancel{12} \qquad x \geq 0$$

$$0 \le 3x - 2x$$

$$x \geq 0 \qquad 0 \le x$$

17. Which is the solution set of $|x| + 7 = 3$? ϕ

(A) $\{-4\}$ (B) $\{4, -4\}$ (C) $\{\}$ (D) $\{-7, -3\}$

$$|x| = 3-7 = -4$$

$\sqrt{\ } \ne -ve$

$(\ )^2 \ne -ve$

$|\ | \ne -ve$

$$|x| = -4 \quad \times$$

18. Which one is the solution set of $\sqrt{5x} = -10$?

(A) $\{\}$ (B) $\{-20\}$ (C) $\{20\}$ (D) $\{-2\}$

ϕ

Dhi

$$|-5| = 5$$
$$|5| = 5$$

19. Which one is the solution set of $5|x| = 25$?

(A) $\{-25, 25\}$ (B) $\{-25\}$ (C) $\{5\}$ (D) $\{-5, 5\}$

$$|x| = \frac{25}{5} = 5$$

$x = ??!$

$$|x| = 5$$

$$|x| = \begin{cases} +x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

$$|x| = \begin{cases} -x, & x < 0 \end{cases}$$

$$|-5| = 5$$
$$|5| = 5$$

20. Which one is the solution set of $\sqrt{3x+1} = 5$?

(A) 25 (B) 8 (C) 24 (D) $\frac{26}{3}$

$$\left(\sqrt{3x+1}\right)^2 = (5)^2$$
$$3x+1 = 25$$
$$\frac{3x}{3} = \frac{24}{3}$$
$$x = 8$$

Section B - Short Questions 60 Marks

Q.2 Solve the following questions. Each question carries 4 marks.

(i) Solve the inequality and plot the solution on the real number line: **[4 Marks]**

$$-3x + 11 \geq 5(x+7) - 32 \geq 3x - 1, \qquad x \in \mathbb{R}$$

$$\Rightarrow -3x+11 \geq 5x+35-32 \geq 3x-1$$
$$\Rightarrow -3x+11 \geq 5x+3 \geq 3x-1$$
$$-3x+11 \geq 5x+3 \quad \text{and} \quad 5x+3 \geq 3x-1$$
$$11-3 \geq 5x+3x \quad \text{and} \quad 5x-3x \geq -1-3$$

$11 - 3 \geq 5x + 3x$ and $5x - 3x \geq -1 - 3$

$8 \geq 8x$ and $2x \geq -4$

$\Rightarrow 8x \leq 8$ and $2x \geq -4$

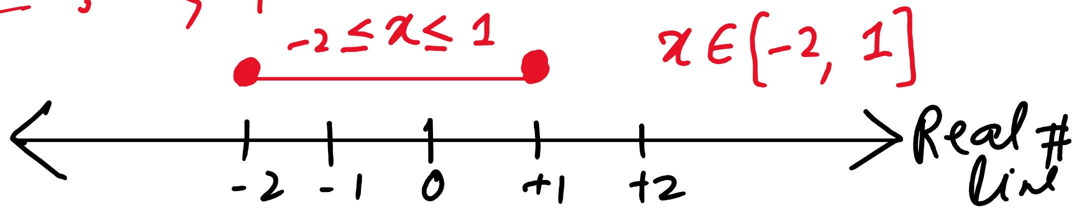
$\Rightarrow x \leq \frac{8}{8}$ and $x \geq -\frac{4}{2}$

$\Rightarrow \boxed{x \leq 1}$ and $\boxed{x \geq -2}$

S.S: $\{x \mid x \in \mathbb{R} \wedge x \leq 1\}$ and $\{x \mid x \in \mathbb{R} \wedge x \geq -2\}$

Solution set: $\{x \mid x \in \mathbb{R} \wedge -2 \leq x \leq 1\}$ Ans

$x \in [-2, 1]$



(ii) Solve the following inequalities: **[4 Marks]**

$$2x - 3 < x + 1 \quad \text{and} \quad 3x - 3 \geq 2x + 8$$

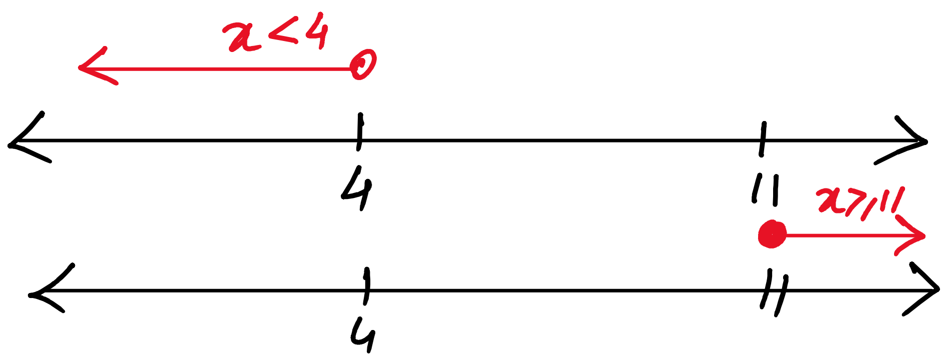
Sol:-

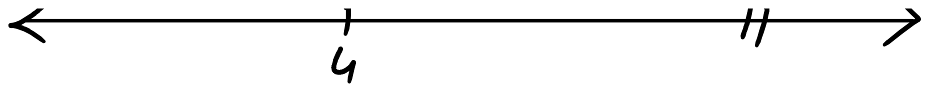
$\Rightarrow 2x - x < 3 + 1$ and $3x - 2x \geq 8 + 3$

$\Rightarrow x < 4$ and $x \geq 11$

$4 > x \geq 11$

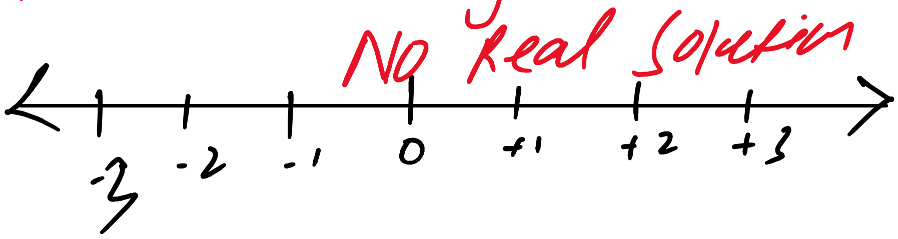
S.S $= \{x \mid x \in \mathbb{R} \wedge 4 > x \geq 11\} = \{\ \} = \phi$





Hence S.S = { } or ϕ because there is NO intersection (common points) b/w these two compound inequalities.

Hence NO shade region on Real No line.



(iii) Solve the inequality and plot the solution on the real number line: **[4 Marks]**

$$3x + 5 \leq 5 - 3(x+2) \leq 6x - 10, \qquad x \in \mathbb{R}$$

Sol:-

$\Rightarrow \quad 3x+5 \leq 5-3x-6 \leq 6x-10$

$\Rightarrow \quad 3x+5 \leq -3x-1 \leq 6x-10$

$\Rightarrow \quad 3x+5 \leq -3x-1$ and $-3x-1 \leq 6x-10$

$\Rightarrow \quad 6x \leq -5-1$ and $-1+10 \leq 6x+3x$

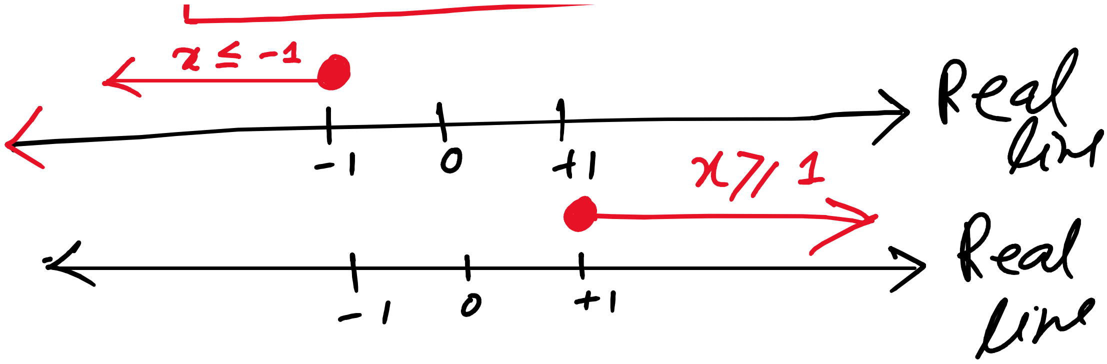
$\Rightarrow \quad 6x \leq -6$ and $9 \leq 9x$

$\Rightarrow \quad \boxed{x \leq -1}$ and $1 \leq x$

$\Rightarrow \quad \boxed{x \leq -1}$ and $\boxed{x \geq 1}$

$\Rightarrow \quad \boxed{-1 \geq x \geq 1}$

Real

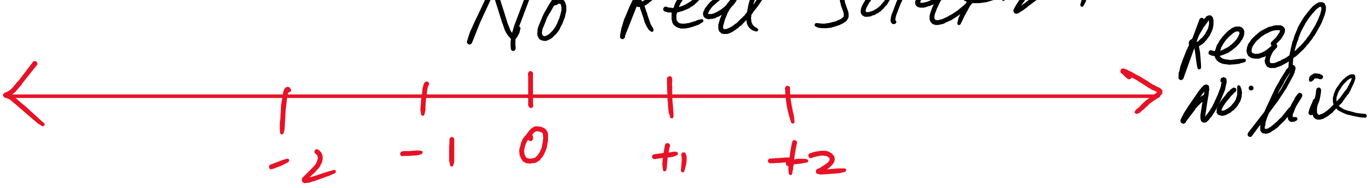


As there is NO common points i.e. intersection b/w two inequalities

So, hence $S.S: \{x \mid x \in \mathbb{R} \wedge -1 \geq, x \geq 1\}$

On Real No. line: $S.S = \{\ \}$ or $\emptyset$ (Phi)

No Real Solution



(iv) Solve the linear equation: **[4 Marks]**

$$\frac{1}{3}(x-2) + \frac{2-3x}{2} = \frac{x+5}{6}$$

Sol:

Multiply both sides with 6 (LCM)

2	2 - 3 - 6
3	1 - 3 - 3
	1 - 1 - 1

$$2(x-2) + 3(2-3x) = x+5$$

$$\Rightarrow 2x - 4 + 6 - 9x = x + 5$$

$\Rightarrow 2x - 9x + 5 = \ldots$

$2x - 9x - x = 5 - 2$

$-8x = 3$

$\boxed{x = -\frac{3}{8}}$ Ans

(v) Solve the given linear equation and represent the solution on a real number line:

$$\frac{5}{6}x - \frac{7}{3} = \frac{3}{4}x + \frac{5}{2} \rightarrow (1)$$

2	2-3-4-6
2	1-3-2-3
3	1-3-1-3
	1-1-1-1

$\times$ by 12 (LCM) on both sides, we get of Eq. (1)

$\frac{7}{3} \times 12$

$\frac{3}{4} \times 12$

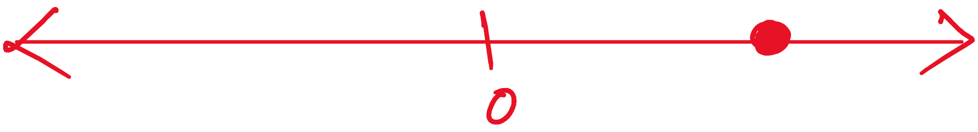
$\Rightarrow 2 \times 5x - 7 \times 4 = 3x \times 3 + 5 \times 6$

$\Rightarrow 10x - 28 = 9x + 30$

$10x - 9x = 30 + 28$

$\boxed{x = 58}$ Ans

$x = 58$



(vi) Solve the equation and represent the solution on a real number line:

Sol:

$$\frac{3x-5}{4} = \frac{2x+1}{3}$$

By cross-multiply, we get

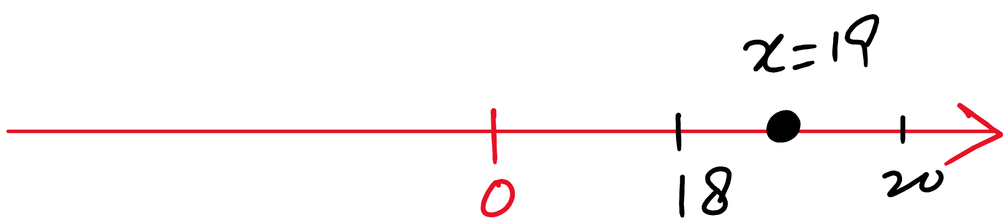
$$3(3x-5) = 4(2x+1)$$

$$\Rightarrow \quad 9x - 15 = 8x + 4$$

$$9x - 8x = 15 + 4$$

$$\boxed{x = 19} \quad \text{Ans}$$

✓



(vii) Solve: **[4 Marks]**

$$\frac{3x}{5} - \frac{1}{2} = \frac{x}{4} + 1 \rightarrow (1)$$

× by 20(LCM) on b.s of Eq. (1), we get

x by 20(LCM) on b.s of Eqn ①, we get

$$\Rightarrow 4(3x) - 10 = 5x + 20$$

$$\Rightarrow 12x - 10 = 5x + 20$$

$$12x - 5x = 10 + 20$$

$$7x = 30$$

$$\boxed{x = \frac{30}{7}}$$ A

(viii) Solve: **[4 Marks]**

$$\frac{2x+3}{5} = \frac{3-4x}{8}$$

Sol:

$$8(2x+3) = 5(3-4x)$$

$$\Rightarrow 16x + 24 = 15 - 20x$$

$$16x + 20x = 15 - 24$$

$$36x = -9$$

$$x = -\frac{9}{36}$$

$$\boxed{x = -\frac{1}{4}}$$ A

(ix) Solve: **[4 Marks]**

$$\frac{4(x+2)}{3} - \frac{6(x-7)}{7} = 12$$

Sol:

X by 21 on both sides, we get

$$7 \times 4(x+2) - 3 \times 6(x-7) = 252$$

$$28(x+2) - 18(x-7) = 252$$

$$28x + 56 - 18x + 126 = 252$$

$$10x = 252 - 182$$

$$\Rightarrow 10x = 70$$

$$= 79$$

x

$\boxed{x = 7}$ 1

(x) Solve: **[4 Marks]**

$$\frac{1}{5}(x-8)+\frac{4+x}{7}=7-\frac{23-x}{5}$$

Sol:-

$\times$ by 35(LCM) on B.S, we get

$$7(x-8)+5(4+x)=245-7(23-x)$$

$$\cancel{7x}-56+20+5x=245-161+\cancel{7x}$$

$$5x = 245-161+36$$

$$5x = 120$$

$$x = \frac{120}{5}$$

$$\boxed{x = 24} \text{ Ans}$$

Example 12: Solve the following inequality.

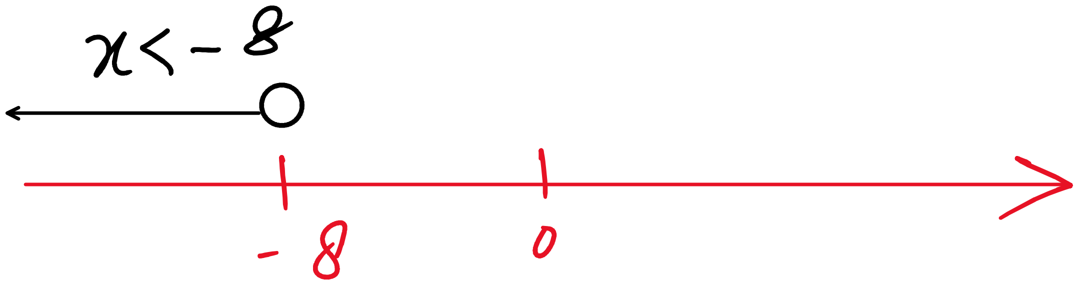
$8x + 9 < 6x - 7$, where $x \in R$.

$$\Rightarrow \quad 8x - 6x < -9 - 7$$

$$\Rightarrow \quad 2x < -16$$

$$x < \frac{-16}{2}$$

$$\boxed{x < -8}$$



The solution is $\{x \mid x \in R \wedge x < -8\}$

Example 13: Solve the inequality.

$3(-4 + 5y) \leq -8(1 - 2y) + 6$, ~~x~~ y $\in$ R

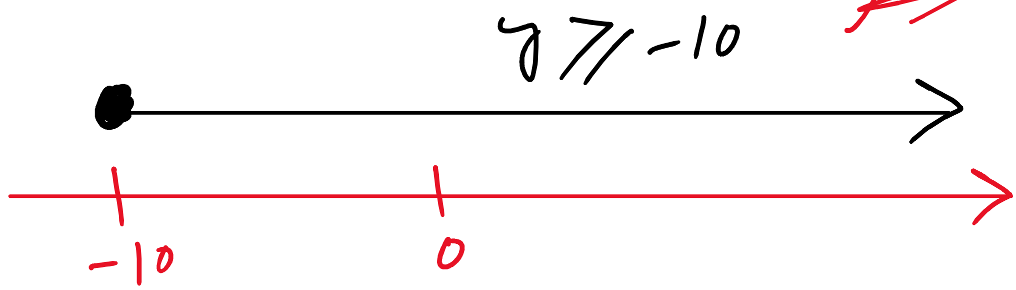
$$\Rightarrow \quad -12 + 15y \leq -8 + 16y + 6$$

$$15y - 16y \le -8 + 6 + 12$$

$$-y \le 10$$

$$\boxed{y \ge -10}$$

$$S.S = \{ y \mid y \in R \wedge y \ge -10 \}$$



Z → Integers

$$\frac{1}{2}x + 3 \ge \frac{1}{4}x + 2, x \in R$$

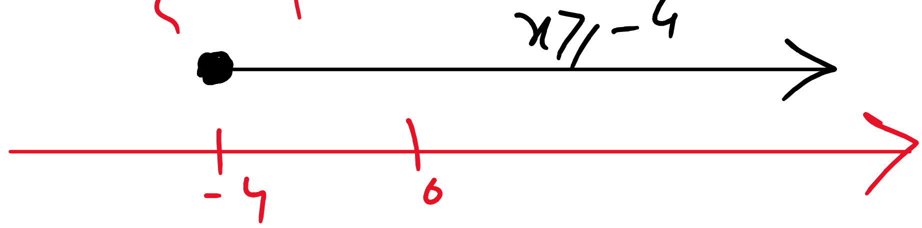
× by 4 on B.S; we get.

$$2x + 12 \ge x + 8$$

$$2x - x \ge 8 - 12$$

$$\boxed{x \geq -4}$$

S.S.: $\{x \mid x \in R \wedge x \geq -4\}$



Example 15: Solve the following.

$2x + 3 > 7$ or $4x - 1 < 3$; $x \in R$

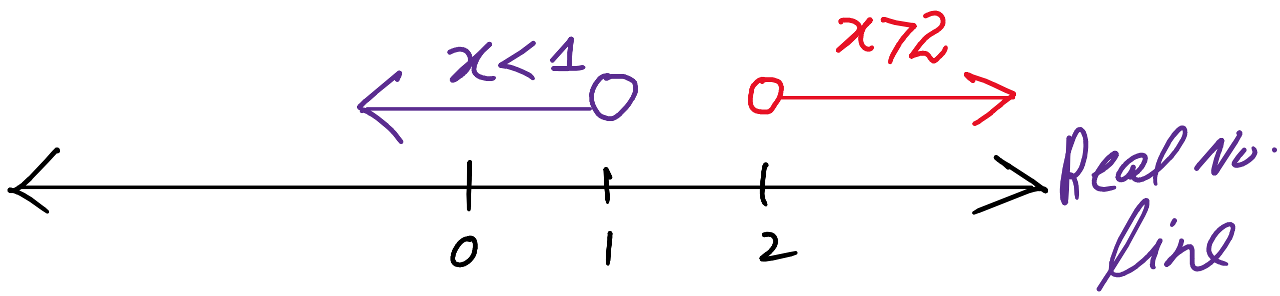
Sol:

$\Rightarrow 2x > 7 - 3$ or $4x < 1 + 3$

$\Rightarrow 2x > 4$ or $4x < 4$

$\Rightarrow x > 2$ or $x < 1$

S.S. $= \{x \mid x \in R \wedge x > 2 \text{ or } x < 1\}$



and

$x-2 < 2x+3 \leq -2x+5$

$$\boxed{5x-2 \le 2x+3 \le -2x+5}$$

Example 16: Solve the compound inequality.
$x - 5 \geq -1$ and $x + 3 \leq 10, x \in \mathrm{R}$

$x \geq 5-1$ and $x \leq 10-3$

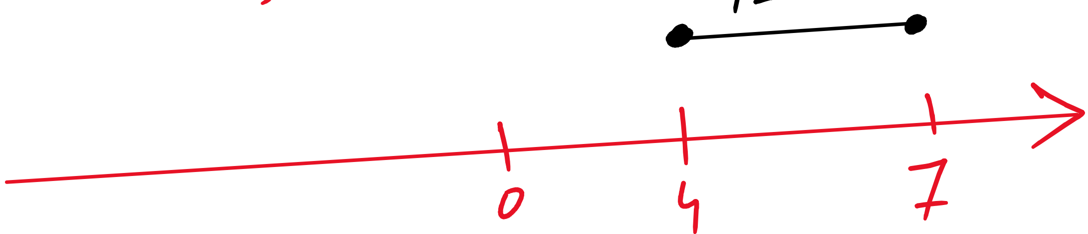
$x \geq 4$ and $x \leq 7$

and

S.S: $\{x \mid x \in R \wedge x \geq 4\}$ $\boxed{4 \leq x \leq 7}$ $\{x \mid x \in R \wedge x \leq 7\}$

$S.S = \{x \mid x \in R \wedge 4 \leq x \leq 7\}$

$4 \leq x \leq 7$



$x \in [4, 7]$

1st $2x$ 2nd $(5, 17)$

Example 17: The sum of two times a number x and 3 is between 5 and 17. Between what two numbers is the given number x?

Sol.

Sol. 2

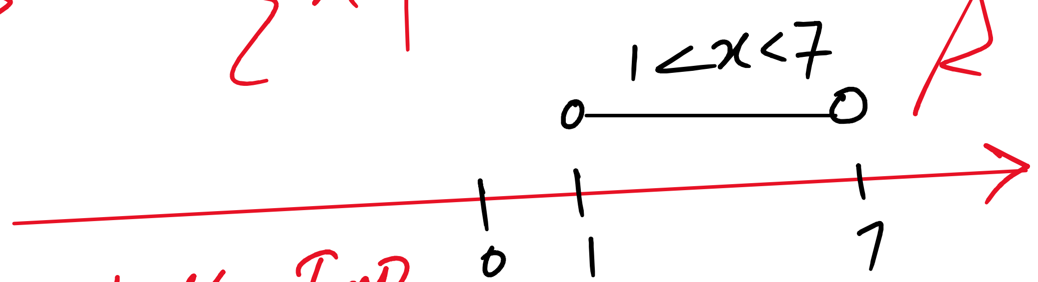
$$5 < 2x+3 < 17$$

$2x+3 > 5$	$2x+3 < 17$
$2x > 5-3$	$2x < 17-3$
$2x > 2$	$2x < 14$
$\boxed{x > 1}$	$x < 7$

$$\boxed{1 < x < 7}$$

$$S.S = \{x \mid x \in R \wedge 1 < x < 7\}$$

R



Less Imp

(xi) Solve: **[4 Marks]**

$$4 + 2\sqrt{3y+1} = 3$$

$$2\sqrt{3y+1} = 3-4$$

$$2\sqrt{3y+1} = -1$$

$$\sqrt{3y+1} = -\frac{1}{2} < 0$$

$$S.S = \{ \} \text{ or } \phi$$

Because square root of any Real No is always non-negative

Not summative Topic

(xii) Solve: **[4 Marks]**

$$\sqrt{\frac{a+6}{a+2}} = \sqrt{\frac{a+2}{a-1}}$$

Sol: squaring both sides, we get

$$\frac{a+6}{a+2} = \frac{a+2}{a-1}$$

$$(a+6)(a-1) = (a+2)^2$$

$$a^2 - a + 6a - 6 = a^2 + 4 + 2(a)(2)$$

$$\cancel{a^2} + 5a - 6 = \cancel{a^2} + 4 + 4a$$

$$\cancel{a^2}+5a-6=\cancel{a^2}+4+4a$$

$$5a-4a=4+6$$

$$\boxed{a=10}\ A$$

Thus $s.s=\{10\}$ A

(xiii) Solve: **[4 Marks]**

$$\sqrt{9-2x}=\sqrt{5x-12}$$

↳ Not Summative

Sol: Squaring both sides, we;

$$9-2x=5x-12$$

$$-2x-5x=-9-12$$

$$\cancel{-}7x=\cancel{-}21$$

$$21$$

$$x = \frac{21}{7}$$

$$\boxed{x = 3}$$

On checking, 3 satisfies the original radical eqn.

Hence $S.S = \{3\}$ A

(xiv) Solve, where $x \in \mathbb{R}$: **[4 Marks]**

$$|2x + 5| = 11$$

Not summative

$$\Rightarrow \quad 2x + 5 = \pm 11$$

$2x + 5 = 11$ or $2x + 5 = -11$

$2x = 11 - 5$ | $2x = -11 - 5$

$2x = 6$ | $2x = -16$

$\boxed{x = 3}$ or $\boxed{x = -8}$

Thus, $S.S. = \{3, -8\}$ A

$$|x| = \begin{cases} +x, & x > 0 \\ -x, & x < 0 \end{cases}$$

(xv) Solve the absolute value equation, where $x \in \mathbb{R}$: **[4 Marks]**

Sol → NA Summative

$$\frac{|10-x|}{5} = \frac{|2x-5|}{2}$$

$$2|10-x| = 5|2x-5|$$

$$|20-2x| = |10x-25|$$

$$20-2x = 10x-25$$

$$-2x-10x = -20-25$$

$$-12x = -45$$

$$x = \frac{45}{12}$$

$$\boxed{x = \frac{15}{4}}$$

or

$$20-2x = -(10x-25)$$

$$20-2x = -10x+25$$

$$-2x+10x = 25-20$$

$$8x = 5$$

$$\boxed{x = \frac{5}{8}}$$

On checking, both $x = 15/4$ & $x = 5/8$ satisfy the original absolute value Eqn. So

$$S.S = \left\{ \frac{5}{8}, \frac{15}{4} \right\}$$

End of Chapter